Priyanka Giri



AI Engineer | Full-Stack Software Engineer
017657783896 | priyanka.giri.0397@gmail.com
LinkedIn: https://www.linkedin.com/in/priyanka-giri-fullstack-developer/
GitHub: https://github.com/Pikuz1
Portfolio: https://priyanka-giri-fullstack-developer.netlify.app/

Summary

Software Engineer with 5+ years of experience building production-grade web platforms across fintech, analytics, and automation, now focused on AI engineering. Builds **LLM** applications end to end: retrieval-augmented generation (**RAG**), embeddings and vector search, and conversational chatbot interfaces, with **Python/FastAPI** backends and **React/TypeScript** frontends. Strong command of **REST API** integration, data visualization, testing, and **CI/CD**, with academic background in machine learning and reinforcement learning.

Core Skills

AI & LLM: RAG, LangChain, Embeddings (sentence-transformers), Vector Databases (ChromaDB), Gemini API, Prompt Engineering, Retrieval Evaluation
Frontend: React, Next.js, Angular, TypeScript, Tailwind, CSS, SCSS, Chart.js, Three.js
State Management: RxJS, NgRx, React Context, custom hooks
Testing & Quality: Playwright (E2E), pytest, Vitest, Jest, Testing Library, Cypress
APIs & Real-time: RESTful APIs, WebSocket, async/await, error handling
Backend: Python, FastAPI, Java, Node.js, Flask, SQL, SQLite, MongoDB, Firebase
Cloud & DevOps: AWS (S3, Lambda), Docker, GitHub Actions, CI/CD
Tooling: Git, Storybook, Figma, Jira, Clickup
Concepts: UI architecture, design systems, component libraries, accessibility (a11y), data modeling

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Experience

Fullstack Engineer - Early-Stage Startup Product Oct. 2025 - Jun. 2026

LLM-Powered OLAP Analytics Platform *Hamburg, Germany*

- Architected and built a full-stack **LLM-powered analytics** platform from scratch, owning frontend systems end-to-end.
- Integrated the backend's natural-language-to-SQL service into the conversational **chatbot UI** via REST APIs, using response flags to render answers as interactive **Chart.js** charts users can explore.
- Designed and implemented interactive dashboards with complex data visualizations for **OLAP** workloads.
- Developed frontend architecture, services, and data models with a focus on scalability, consistency, and reliability.
- Collaborated with founders on system design, **MVP scope**, and technical strategy.

Software Engineer - Specter Automation May. 2022 - Oct. 2024

Construction Automation Platform (B2B) *Cologne, Germany*

- Developed and maintained a large-scale **Angular + TypeScript** B2B platform with complex UI/UX workflows.
- Built **3D visualization systems** using **Three.js,** delivering interactive spatial interfaces for construction use cases.
- Introduced **end-to-end testing** with **Playwright/Cypress** and contributed to **CI/CD** pipelines via GitHub Actions.
- Improved frontend quality through systematic debugging, performance optimization, and code reviews.

Junior Frontend Developer - Coindex Dec. 2019 - Aug. 2021

Cryptocurrency platform *Bielefeld, Germany*

- Built **React + TypeScript** dashboards for real-time cryptocurrency data with **REST API** and **WebSocket** integrations.
- Developed reusable component libraries and data visualization features for live market analytics.
- Contributed to **testing automation** and **AWS deployments** (S3, Lambda).

ACADEMIC & TECHNICAL PROJECTS

DocuMind: Multilingual RAG Assistant for Contracts

Personal Project Oct. 2026

- Built a full-stack **RAG application**: upload a German or English contract PDF, ask questions, and get answers citing the source page (**Python**, FastAPI, **LangChain**, ChromaDB, Gemini, React, **TypeScript**).
- Built the ingestion pipeline with page-aware chunking and local multilingual embeddings; evaluated retrieval with a golden question set and tuned chunking to reach **100% hit@4**.
- Privacy-first and production-ready: uploads deleted after indexing, 190+ tests with 100% coverage, Playwright E2E, **Docker**, and **GitHub Actions CI** on Linux, Windows, and macOS.

Near-Shortest Path Routing in Hybrid Communication Networks (Master's Thesis)

Paderborn University Feb. 2025 - Jul. 2025

- Designed and implemented routing algorithms in **Python** using **NetworkX** and **Matplotlib**.
- Focused on scalable routing strategies in hybrid network models.

AICON: Artificial Intelligence for Computer Networks (Student Assistant)

Paderborn University Apr.2020 - Mar. 2021

- Routing of the packets in a wireless network to their respective destination.
- Analysis of Reinforcement Learning algorithms in wireless networks using **Python**.
- Minimizing the packet loss due to important challenges such as **signal interference**, **defect nodes** in the network, and **hidden terminal problem**.

Car Damage Assessment System

Mumbai University Jun. 2018 - May 2019

- Built a machine-learning-based image classification system for insurance use cases.
- Developed frontend design using **Angular 6**.
- Co-authored a paper on machine-learning-based damage assessment for insurance

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Education

Paderborn University Oct. 2019 - Jan. 2026

Master's in Computer Science Paderborn, Germany

Mumbai University Jun. 2015 - May 2019

Computer Engineering Mumbai, India

Language Skills

English: C1 (Fluent - Professional Proficiency)

German: B1 (Intermediate - in progress)